

ASSESSMENT TOOL 1

COURSE CODE: CSA0713 COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Sivadharshini.K(192524008)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

Signature of the student:sivadharshini.k

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

i.) Use polynomial long division to determine the message that should be transmitted.

ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission

link. What is the result of the receiver's CRC calculation? How does the receiver know that

an error has occurred?

Given:

Message = 11100011

Generator Polynomial = $x^3 + 1$

Binary representation of polynomial = 1001

i) Determine the message to be transmitted

Step 1: Append zeros

Since the polynomial degree is 3, append 3 zeros to the message.

Original message:

11100011

After appending 3 zeros:

11100011000

Step 2: Perform Modulo-2 (XOR) Division

Divide:

$11100011000 \div 1001$

Using polynomial (mod-2) division, the remainder obtained is:

100

Step 3: Form the transmitted message

Replace the appended zeros with the remainder.

Original message:

11100011

CRC bits:

100

Transmitted message:

11100011100

Answer (i):

The transmitted message is 11100011100.

ii) Leftmost bit is inverted during transmission

The transmitted message is:

11100011100

If the leftmost bit is inverted, it becomes:

01100011100

The receiver divides the received data by the same generator polynomial (1001).

CRC Check

$01100011100 \div 1001$

Remainder obtained:

101

Since the remainder is not 000, the CRC check fails.

How does the receiver know an error has occurred?

In CRC, the received data should produce a remainder of 000 if no error has occurred.

Here, the remainder is 101 (non-zero).

Therefore, the receiver concludes that the data has been corrupted during transmission and rejects the frame or requests retransmission.

Final Answer

(i)

Generator Polynomial = 1001

Appended message = 11100011000

CRC Remainder = 100

Transmitted message = 11100011100

(ii)

Received message after leftmost bit inversion = 01100011100

CRC remainder at receiver = 101

Since the remainder is not 000, the receiver detects that an error has occurred and rejects the received message.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this

network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is 3162. Find the channel capacity

Given:

(a) Throughput Calculation

Given:

- Bandwidth = **10 Mbps**
- Frames per minute = **12,000**
- Bits per frame = **10,000 bits**

Formula

$$\text{Throughput} = \frac{\text{Frames per minute} \times \text{Bits per frame}}{60}$$
$$\text{Throughput} = \frac{12000 \times 10000}{60} = 2000000 \text{ bits/s} = 2 \text{ Mbps}$$

Calculation

$$\text{Throughput} = \frac{12000 \times 10000}{60}$$
$$\text{Throughput} = 2000000 \text{ bits/s} = 2 \text{ Mbps}$$

Answer (a):

Throughput = 2 Mbps

(b) Channel Capacity (Shannon's Formula)

Given:

- Bandwidth, $B = 3000 \text{ Hz}$
- Signal-to-Noise Ratio (SNR) = **3162**

Formula

$$C = B \log_2(1 + \text{SNR})$$

Calculation

$$C = 3000 \times \log_2(1 + 3162)$$
$$C = 3000 \times \log_2(3163) = 3000 \times 11.64 = 34920 \text{ bits/s}$$

Using,

$\log_2(3163) \approx 11.63$ $\log_2(3163) \approx 11.63$ $C = 3000 \times 11.63$ $C = 3000 \times 11.63$ $C \approx 34,890$ bps $C \approx 34,890$ bps $C \approx 34.9$ kbps

Answer (b):

Channel Capacity ≈ 34.9 kbps ($\approx 34,890$ bps)

Final Answers

a) Throughput = 2 Mbps

b) Channel Capacity = 34.9 kbps ($\approx 34,890$ bps)

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the Hamming (12,8) error-correcting code.

Given:

- Wireless patient-monitoring devices send **8-bit patient data**.
- Noise may change **one bit** during transmission.
- The hospital uses **Hamming (12,8)** code to detect and correct a single-bit error.

Encoding Process

1. The original **8-bit data** is taken.
2. Four parity bits (**P1, P2, P4, P8**) are added, making a **12-bit codeword**.
3. The 12-bit codeword is transmitted over the wireless channel.

Error During Transmission

- Due to electromagnetic interference, one bit may be flipped ($0 \rightarrow 1$ or $1 \rightarrow 0$).

Error Detection

1. At the receiver, the parity bits are checked.
2. A **syndrome** is calculated using the parity checks.
3. If the syndrome is **0000**, there is **no error**.
4. If the syndrome is **non-zero**, it indicates the **exact position** of the erroneous bit.

Error Correction

- The receiver flips the incorrect bit identified by the syndrome.
- The original **8-bit patient data** is recovered automatically.
- No retransmission is required.

Advantages in the Hospital

- ✓ Detects and corrects **single-bit errors**.
- ✓ Ensures accurate heart rate and blood pressure data.
- ✓ Reduces delay because retransmission is not needed.
- ✓ Improves reliability of wireless patient monitoring.
- ✓ Enables doctors to receive correct patient information in real time.

Conclusion

Using the **Hamming (12,8) error-correcting code**, the hospital can automatically detect and correct any **single-bit transmission error**, ensuring reliable and continuous monitoring of patients without waiting for the data to be sent again.

4. a) A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Given

- Network Address: **172.16.10.0/24**
- Subnet Mask: **/27 (255.255.255.224)**

Step 1: Calculate the Number of Subnets

Borrowed bits:

$$27 - 24 = 3 \text{ bits}$$

Number of subnets:

$$2^3 = 8$$

Answer: 8 subnets

Step 2: Calculate Usable Hosts per Subnet

Host bits:

$$32 - 27 = 5 \quad 2^5 = 32$$

Total addresses per subnet:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30$$

Answer: 30 usable hosts per subnet

Step 3: List the /27 Subnets

Subnet	Network Address	Host Range	Broadcast Address
1	172.16.10.0	172.16.10.1 – 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 – 172.16.10.62	172.16.10.63
3	172.16.10.64	172.16.10.65 – 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 – 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 – 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 – 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 – 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 – 172.16.10.254	172.16.10.255

Step 4: Determine the Subnet of 172.16.10.78

The IP **172.16.10.78** lies between **172.16.10.65** and **172.16.10.94**.

Therefore,

- **Subnet:** 172.16.10.64/27
- **Network Address:** **172.16.10.64**
- **Broadcast Address:** **172.16.10.95**

Step 5: Check Whether 172.16.10.95 Is in the Same Subnet

- **172.16.10.95** is the **broadcast address** of the subnet.
- It belongs to the same subnet but **cannot be assigned to a host**.

Answer: Yes, it is within the same subnet range, but it is the **broadcast address**, not a usable host address.

4(b) IPv6 Addressing

Given IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the Address

Rules:

- Remove leading zeros.
- Replace consecutive groups of zeros with "::" (only once).

Compressed Address:

2001:db8::ff00:42:8329

Step 2: Expand the Address

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix (/64)

With a /64 prefix, the first **64 bits** (first four groups) represent the network.

Network Prefix:

2001:0db8:0000:0000::/64

or simply

2001:db8::/64

Step 4: Calculate the Number of Host Addresses

Host bits:

$$128 - 64 = 64$$

Number of host addresses:

$$2^{64} = 18,446,744,073,709,551,616$$

Final Answers

4(a)

- Number of subnets = **8**
- Usable hosts per subnet = **30**
- IP **172.16.10.78** belongs to subnet **172.16.10.64/27**
- Network address = **172.16.10.64**
- Broadcast address = **172.16.10.95**
- IP **172.16.10.95** is the **broadcast address** of the same subnet and **cannot be assigned to a host**.

4(b)

- Compressed IPv6 address = **2001:db8::ff00:42:8329**
 - Expanded IPv6 address = **2001:0db8:0000:0000:ff00:0042:8329**
 - Network prefix (/64) = **2001:db8::/64**
 - Total possible host addresses = $2^{64} = 18,446,744,073,709,551,616$ (≈ 18.4 quintillion)
5. A university is conducting an online semester examination for 8,000 students simultaneously. At exactly 10:00 AM, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a 10 Gbps high-speed network, while students access the examination portal using different internet connections such as fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps). During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. Analyze the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

Primary Networking Problem

The primary networking problem is **Flow Control**.

Flow control ensures that the **sender does not transmit data faster than the receiver can receive and process it**. In this scenario, the university server sends encrypted question papers at a very high speed (10 Gbps), while many students have much slower internet connections (100 Mbps, 20 Mbps, or 5–10 Mbps).

How the Speed Mismatch Affects Communication

- The examination server transmits data much faster than some student devices can receive.

- Students with slower internet connections cannot process incoming packets quickly.
- The receiver's **buffer becomes full (buffer overflow)**.
- Some packets are dropped because there is no space in the buffer.
- Lost packets require **retransmission**, increasing network traffic.
- Delayed acknowledgments (ACKs) cause the sender to resend data unnecessarily.
- Students experience incomplete downloads, temporary freezing, and delays in receiving the question papers.
- Overall network performance and reliability decrease.

Role of Flow Control

Flow control regulates the sender's transmission rate based on the receiver's capacity. Protocols such as **TCP** use mechanisms like the **sliding window** to adjust the amount of data sent before waiting for acknowledgments. This prevents buffer overflow and improves reliable communication.

Effects Observed in the Scenario

- Buffer overflow at receivers.
- Packet loss.
- Increased retransmissions.
- Delayed acknowledgments.
- Network congestion.
- Slow and incomplete download of question papers.
- Temporary freezing of the examination portal.

Conclusion

The problem is caused by a **mismatch between the sender's transmission speed and the receiver's processing capability**. By using **flow control**, the sender adjusts its transmission rate according to the receiver's capacity, preventing packet loss, reducing retransmissions, avoiding buffer overflow, and ensuring reliable and efficient delivery of the examination question papers.